

Machine Learning-Specific Code Smell Detection

Summer Research Internship 2026

National Institute of Technology Calicut (NIT Calicut)

Project Title: Machine Learning-Specific Code Smell Detection

Duration: 15 May 2026 – 15 July 2026

Department: Computer Science and Engineering

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About the Internship

This document provides a public overview of my Summer Research Internship at the National Institute of Technology Calicut. The internship focused on understanding Machine Learning-specific code smells, software quality, dataset engineering, verification, documentation, and research methodology. Confidential implementation details and datasets are intentionally excluded.

Project Objectives

- Understand Machine Learning-specific code smells.
- Study their impact on software quality.
- Support dataset engineering activities.
- Participate in verification.
- Learn refactoring concepts.
- Contribute to research documentation.
- Gain practical research experience.

Research Workflow

Research Problem

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Literature Review

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Understanding ML Code Smells

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Dataset Engineering

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Dataset Verification

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Code Refactoring

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Documentation

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Final Deliverables

Technologies Used

- Python
- Git
- GitHub
- JSON

- AST
- Google Colab
- VS Code
- LLMs

Research Contributions

Contributed to literature review, dataset engineering activities, verification, code quality analysis, documentation, and understanding of research-oriented software engineering practices.

Skills Developed

- Python Programming
- Git & GitHub
- Research Methodology
- Dataset Verification
- Code Refactoring
- Technical Documentation
- Software Quality Analysis

Internship Outcomes

The internship strengthened my understanding of collaborative research, documentation, version control, software quality, and structured problem solving.

Future Scope

- Explore more ML-specific code smells.
- Improve automated detection.
- Investigate AI-assisted software quality analysis.
- Expand datasets to multiple languages.

Acknowledgement

I sincerely thank Dr. Shweta and the Department of Computer Science and Engineering, National Institute of Technology Calicut, for their guidance and support throughout the internship.

Disclaimer

This document is intended only as a public showcase of my internship experience.
Confidential research materials, datasets, and implementation details are not included.